

Benchmark results compiled with -O2: mean time (microseconds)

Test case	heftia	freer	polysemy	fused	effectful	eff	effective	effective.fstg	effective.naive	mp	mp.safe	mtl.logict	handrolled.logict	effective.lstg	mtl.lisst	handrolled.lisst
countdown.shallow.10000	162.509	148.036	1254.828	4.649	145.66	280.929	268.414	2.266	88.74	188.439	1045.589	4.558	4.549	-	-	-
countdown.deep.10000	1283.645	571.216	2314.552	2664.256	144.508	279.122	162.059	2.328	168.719	193.497	3478.359	4.554	4.555	65.095	-	-
countdown.very-deep.10000	9253.212	5055.197	11067.506	-	163.653	276.022	168.594	2.332	1309.456	522.355	22834.638	4.552	4.552	67.057	-	-
catch.shallow.10000	273.135	32.787	1116.568	20.484	1066.619	396173	247.752	16.139	73.291	841.964	834.509	20.422	20.427	-	-	-
catch.deep.10000	3236.105	35.125	6534.984	1712.121	1072.883	396733	176.323	15.874	238.262	843.314	848.246	20.37	20.251	166.434	-	-
catch.very-deep.10000	40833.875	34.942	40599.075	-	1206.105	378904.7	176.058	16.182	1814.159	862.115	854.228	20.355	20.223	164.374	-	-
local.shallow.10000	915.613	454.254	2212.628	4.932	539.943	397097.4	195.454	2.491	136.943	787.702	787.475	4.955	4.925	-	-	-
local.deep.10000	3891.412	450.111	5065.709	4.863	537.271	396393.6	199.373	2.467	220.053	786.133	782.384	4.803	4.806	199.466	-	-
local.very-deep.10000	36142.113	444.232	28630.162	-	568.941	399390.6	199.71	2.484	880.395	793.056	783.554	3514.791	3509.508	199.093	-	-
nondet.shallow.32	2043.39	3358.445	8380.572	240.693	-	24548.775	760.461	79.85	908.047	1753.266	1751.654	221.142	243.134	-	366.176	345.549
nondet.deep.32	5425.503	5651.225	8073.2	2468.494	-	49823.3	752.38	79.254	602.678	5612.009	5627.837	241.616	242.787	989.093	62546.925	62749.525
nondet.very-deep.32	41866.25	26123.362	7743.666	-	-	387499.4	750.515	79.889	631.973	38866.7	38361.2	248.927	246.92	1006.255	672748.2	672011.6

Benchmark results compiled with -O2: mean time (% of fastest)

Test case	heftia	freer	polysemy	fused	effectful	eff	effective	effective.fstg	effective.naive	mp	mp.safe	mtl.logict	handrolled.logict	effective.lstg	mtl.lisst	handrolled.lisst
countdown.shallow.10000	7170.3%	6531.7%	55366.3%	205.1%	6426.9%	12395.3%	11843.1%	100.0%	3915.4%	8314.4%	46134.1%	201.1%	200.7%	-	-	-
countdown.deep.10000	55133.7%	24534.2%	99412.0%	114432.1%	6206.7%	11988.5%	6060.6%	100.0%	7246.6%	8310.9%	149398.5%	195.6%	195.6%	2795.9%	-	-
countdown.very-deep.10000	396838.6%	216800.1%	474647.4%	-	7018.5%	11837.7%	7230.4%	100.0%	56158.1%	22402.0%	979299.4%	195.2%	195.2%	2875.9%	-	-
catch.shallow.10000	1692.4%	203.1%	6918.4%	126.9%	6608.9%	2454731.2%	1535.1%	100.0%	454.1%	5216.9%	5170.7%	126.5%	126.6%	-	-	-
catch.deep.10000	20385.6%	221.3%	41166.7%	10785.4%	6758.6%	249193.0%	1110.7%	100.0%	1500.9%	5312.4%	5343.5%	128.3%	127.6%	1048.4%	-	-
catch.very-deep.10000	252342.5%	215.9%	250891.5%	-	7453.4%	2341530.2%	1088.0%	100.0%	11211.0%	5327.6%	5278.9%	125.8%	125.0%	1015.8%	-	-
local.shallow.10000	36756.6%	18235.7%	88824.4%	198.0%	21675.6%	1594195.0%	7846.3%	100.0%	5497.5%	31621.8%	31612.6%	198.9%	197.7%	-	-	-
local.deep.10000	157737.1%	18245.1%	205336.9%	197.1%	21778.1%	16067683.2%	8081.5%	100.0%	8919.8%	31865.6%	31713.7%	194.7%	194.8%	8085.3%	-	-
local.very-deep.10000	1455244.9%	17886.8%	115279.4%	-	22908.1%	16081271.5%	8041.2%	100.0%	35448.7%	31932.0%	31549.4%	141521.4%	141308.7%	8016.4%	-	-
nondet.shallow.32	2559.0%	4205.9%	10495.4%	301.4%	-	30743.6%	952.4%	100.0%	1137.2%	2195.7%	2193.7%	276.9%	304.5%	-	458.6%	432.7%
nondet.deep.32	6845.8%	7130.6%	10186.6%	3114.7%	-	62865.7%	949.3%	100.0%	760.4%	7081.1%	7101.1%	304.9%	306.3%	1248.0%	78920.1%	79175.7%
nondet.very-deep.32	52405.3%	32699.4%	9693.0%	-	-	485045.4%	939.4%	100.0%	791.1%	48650.7%	48017.9%	311.6%	309.1%	1259.6%	842100.4%	841178.4%

Benchmark results compiled with -O2: allocated bytes

Test case	heftia	freer	polysemy	fused	effectful	eff	effective	effective.fstg	effective.naive	mp	mp.safe	mtl.logict	handrolled.logict	effective.lstg	mtl.lisst	handrolled.lisst
countdown.shallow.10000	1760153	1600154	13361558	24	1121570	2087322	2480242	24	1040116	3441170	14241126	24	24	-	-	-
countdown.deep.10000	17760413	10400213	37363526	27121286	1136505	2088075	2000194	24	3920274	3442690	41443954	24	24	1200100	-	-
countdown.very-deep.10000	161763327	89600730	253380958	-	1410362	2085107	2000226	104	20401246	3776395	286589390	104	104	1200132	-	-
catch.shallow.10000	4069612	65594	7764971	65586	7846353	413467555	1687794	65585	723953	6081394	6081394	65585	65586	-	-	-
catch.deep.10000	16870476	66058	55153738	16165113	7864097	413406587	1527786	65585	3367874	6083954	6083957	65586	65585	1527786	-	-
catch.very-deep.10000	132080222	70020	371972772	-	8163146	421556535	1527777	65585	18074259	6139781	6139778	65586	65586	1527777	-	-
local.shallow.10000	7205058	4793274	12630078	24	4326222	409642201	1669450	24	1029425	5513578	5513581	24	24	-	-	-
local.deep.10000	16006054	4793733	30232698	24	4341332	410447199	1589474	24	2549554	5515826	5515826	64	64	1589474	-	-
local.very-deep.10000	95215301	4797717	188651432	-	4648698	417736385	1589474	64	9830346	5535986	5535989	28886110	28886110	1589474	-	-
nondet.shallow.32	13454129	15889987	43492610	3606906	-	111261159	1531330	108345	1532762	22646075	22646078	1987330	2003987	-	3664108	3664100
nondet.deep.32	67560386	45646130	43569970	26810714	-	383227040	1531410	108345	1535162	68636521	68636530	1969098	2010676	7408482	469819528	469817235
nondet.very-deep.32	554516984	313450782	44266209	-	-	4657486684	1531410	108345	1559605	483091430	483091428	2035338	2076914	7408483	5480084980	5480085011

Benchmark results compiled with -O2: allocated bytes (% of least allocated)

Test case	heftia	freer	polysemy	fused	effectful	eff	effective	effective.fstg	effective.naive	mp	mp.safe	mtl.logict	handrolled.logict	effective.lstg	mtl.lisst	handrolled.lisst
countdown.shallow.10000	7333970.8%	6667398.3%	55673158.3%	100.0%	4673208.3%	8697175.0%	10334341.7%	100.0%	4333846.7%	14338298.3%	59338025.0%	100.0%	100.0%	-	-	-
countdown.deep.10000	74001720.8%	43334220.8%	155681358.3%	113005358.3%	4735437.5%	8700312.5%	8334141.7%	100.0%	16334475.0%	14344541.7%	172683141.7%	100.0%	100.0%	5000416.7%	-	-
countdown.very-deep.10000	155541660.6%	86154548.1%	243635536.5%	-	1356117.3%	2004910.6%	1923294.2%	100.0%	19616582.7%	3631149.0%	275566721.2%	100.0%	100.0%	1153973.1%	-	-
catch.shallow.10000	6205.1%	100.0%	11839.6%	100.0%	11963.6%	630430.1%	2573.4%	100.0%	1103.8%	9272.5%	9272.5%	100.0%	100.0%	-	-	-
catch.deep.10000	25723.1%	100.7%	84095.0%	24647.6%	11990.7%	631709.6%	2329.5%	100.0%	5135.1%	9276.4%	9276.4%	100.0%	100.0%	2329.5%	-	-
catch.very-deep.10000	201387.9%	106.8%	567161.4%	-	12446.7%	642763.6%	2329.5%	100.0%	27558.5%	9361.6%	9361.6%	100.0%	100.0%	2329.5%	-	-
local.shallow.10000	30021075.0%	19971975.0%	5282525.0%	100.0%	1802925.0%	1708842504.2%	695041.7%	100.0%	4289270.8%	22973241.7%	22973254.2%	100.0%	100.0%	-	-	-
local.deep.10000	66601891.7%	19973887.5%	125967908.3%	100.0%	18088883.3%	1710196602.5%	6622808.3%	100.0%	10623141.7%	22982608.3%	22982608.3%	266.7%	266.7%	6622808.3%	-	-
local.very-deep.10000	148773907.5%	7496432.8%	294767862.5%	-	7263590.6%	652713101.6%	2483553.1%	100.0%	15359915.6%	8649987.1%	8649982.8%	45134546.9%	45134546.9%	2483553.1%	-	-
nondet.shallow.32	12417.9%	14666.1%	40142.7%	3329.1%	-	102691.5%	1413.4%	100.0%	1414.7%	20901.8%	20901.8%	1834.3%	1849.6%	-	3381.9%	3381.9%
nondet.deep.32	62356.7%	42130.4%	40214.1%	24745.7%	-	353709.9%	1413.5%	100.0%	1416.9%	63350.0%	63350.0%	1817.4%	1855.8%	6837.9%	433632.9%	433630.7%
nondet.very-deep.32	511806.7%	289308.0%	40856.7%	-	-	4298755.5%	1413.5%	100.0%	1439.5%	445882.5%	445882.5%	1878.6%	1916.9%	6837.9%	5057995.3%	5057995.3%

Benchmark results compiled with -O0: mean time (microseconds)

Test case	heftia	freer	polysemy	fused	effectful	eff	effective	effective.fstg	effective.naive	mp	mp.safe	mtl.logict	handrolled.logict	effective.lstg	mtl.lisst	handrolled.lisst
countdown.shallow.10000	804.167	506.074	1704.971	1658.439	707.615	859.362	1565.945	170.797	1614.219	834.721	2138.111	639.852	558.017	-	-	-
countdown.deep.10000	2896.558	953.273	2827.777	10465.006	718.164	858.453	1629.166	170.499	3238.637	1412.368	5161.456	5136.594	5709.797	1666.141	-	-
countdown.very-deep.10000	19400.737	5389.619	11802.919	80825.4	788.74	914.677	1682.616	713.212	16346.175	6103.413	28371.75	42262.8	48747.8	1718.625	-	-
catch.shallow.10000	1081.579	304.12	2919.611	3553.045	2130.293	398631.6	958.691	350.879	1060.333	1666.292	1687.483	257.043	222.904	-	-	-
catch.deep.10000	7926.256	303.688	6918.812	15656.725	2132.303	398223.4	1029.005	348.638	3082.713	1656.521	1659.523	3916.697	4201.839	1110.557	-	-
catch.very-deep.10000	78181.15	304.05	39893.375	108437.15	2280.561	402437.2	1051.602	340.907	16685.506	1680.853	1659.134	39044.05	45436.925	1123.949	-	-
local.shallow.10000	4749.562	668.867	2651.328	1433.894	1057.441	397251.8	631.864	298.024	659.618	1410.856	1408.467	353.437	331.618	-	-	-
local.deep.10000	13641.606	671.453	5669.65	5928.328	1066.27	399040	623.402	297.955	1636.48	1411.856	1403.88	1808.484	1601.918	631.916	-	-
local.very-deep.10000	82267.525	789.574	28022.275	37746.6	1138.829	382525.5	633.946	294.49	8922.419	1421.756	1396.139	11981.006	10873.919	642.513	-	-
nondet.shallow.32	7476.019	5798.378	13454.856	2126.384	-	27762	4074.753	557.566	4152.025	2968.061	301.858	2426.183	1948.359	-	7319.35	6826.219
nondet.deep.32	15689.519	7497.387	22821.6	10439.969	-	53417.113	4214.367	5609.491	9717.562	7684.931	7636.975	6669.834	6735.387	4538.675	79408.7	79026.55
nondet.very-deep.32	83482.2	27879.525	98242.35	108425.8	-	392028.8	4346.37	5587.334	53364.8	47173.2	47320.65	48750.8	52449.075	4527.262	732428.6	737889.8

Benchmark results compiled with -O0: allocated bytes

Test case	heftia	freer	polysemy	fused	effectful	eff	effective	effective.fstg	effective.naive	mp	mp.safe	mtl.logict	handrolled.logict	effective.lstg	mtl.listt	handrolled.listt
countdown.shallow.10000	10640947	6080478	17281885	11920971	5202083	6268900	10801445	720162	11601149	7041771	19361627	4080275	2880389	—	—	—
countdown.deep.10000	38643779	14881348	41284587	63365425	5218182	6269552	11213179	720162	15371643	9444764	48965635	30882914	30882963	11365141	—	—
countdown.very-deep.10000	290665273	94088347	257307425	524203812	5500291	6267134	11319005	720153	44261541	31071203	315398037	272104678	282905448	11398616	—	—
catch.shallow.10000	7913916	2571771	22351555	13075339	12013523	416027858	5434003	1462330	6393893	8161829	8161832	1673347	1593389	—	—	—
catch.deep.10000	33116427	2572605	57554635	64806353	12032296	416835560	5845724	1462349	10164211	8165125	8164621	16926443	18046527	5837666	—	—
catch.very-deep.10000	297290508	2579445	374381179	598207168	12340567	424173075	5951194	1462350	39053649	8227053	8221512	177241699	213164871	5871213	—	—
local.shallow.10000	14981955	5771854	14008722	7688341	6726770	411560990	4008485	1767861	4648339	7193877	7193877	2327815	2327869	—	—	—
local.deep.10000	34184540	5772067	31611559	28490763	6742834	412368856	4020204	1767861	6818581	7197165	7196673	9128592	9048645	4012083	—	—
local.very-deep.10000	227160790	5779474	190034506	215712167	7058160	419622564	4125676	1767861	21308169	7226325	7220781	70335376	66655433	4045555	—	—
nondet.shallow.32	53827211	34532243	81644017	18238419	—	131921103	24814529	53824723	29683003	30882026	30882033	17397027	12390872	—	74123187	69116691
nondet.deep.32	137695719	64288499	200681855	63405713	—	403885904	26178323	53824723	42408353	80929299	80928795	43632235	39124643	28604571	543616138	539108738
nondet.very-deep.32	892511668	332096058	1272017977	469909418	—	4678154988	26283795	53824723	139881895	531355803	531350152	279743700	280236058	28636602	4837189475	4837677338

Benchmark results compiled with -O0: allocated bytes (% of least allocated)

Test case	heftia	freer	polysemy	fused	effectful	eff	effective	effective.fstg	effective.naive	mp	mp.safe	mtl.logict	handrolled.logict	effective.lstg	mtl.listt	handrolled.listt
countdown.shallow.10000	1477.6%	844.3%	2399.7%	1655.3%	722.3%	870.5%	1499.9%	100.0%	1610.9%	977.8%	2688.5%	566.6%	400.0%	—	—	—
countdown.deep.10000	5366.0%	2066.4%	5732.7%	8798.8%	724.6%	870.6%	1557.0%	100.0%	2134.5%	1311.5%	6799.3%	4288.3%	4288.3%	1578.1%	—	—
countdown.very-deep.10000	40361.6%	13065.0%	35729.5%	72790.6%	763.8%	870.3%	1571.8%	100.0%	6146.1%	4314.5%	43796.0%	37784.3%	39284.1%	1582.8%	—	—
catch.shallow.10000	541.2%	175.9%	1528.5%	894.1%	821.5%	28449.7%	371.6%	100.0%	437.2%	558.1%	558.1%	114.4%	109.0%	—	—	—
catch.deep.10000	2264.6%	175.9%	3935.8%	4431.7%	822.8%	28504.5%	399.7%	100.0%	695.1%	558.4%	558.3%	1157.5%	1234.1%	399.2%	—	—
catch.very-deep.10000	20329.6%	176.4%	25601.3%	40907.2%	843.9%	29006.3%	407.0%	100.0%	2670.6%	562.6%	562.2%	12120.3%	14576.9%	401.5%	—	—
local.shallow.10000	847.5%	326.5%	792.4%	434.9%	380.5%	23280.2%	226.7%	100.0%	262.9%	406.9%	406.9%	131.7%	131.7%	—	—	—
local.deep.10000	1933.7%	326.5%	1788.1%	1611.6%	381.4%	23325.9%	227.4%	100.0%	385.7%	407.1%	407.1%	516.4%	511.8%	226.9%	—	—
local.very-deep.10000	12849.5%	326.9%	10749.4%	12201.9%	399.2%	23736.2%	233.4%	100.0%	1205.3%	408.8%	408.4%	3978.6%	3770.4%	228.8%	—	—
nondet.shallow.32	434.4%	278.7%	658.9%	147.2%	—	1064.7%	200.3%	434.4%	239.6%	249.2%	249.2%	140.4%	100.0%	—	598.2%	557.8%
nondet.deep.32	526.0%	245.6%	766.6%	242.2%	—	1542.8%	100.0%	205.6%	162.0%	309.1%	309.1%	166.7%	149.5%	109.3%	2076.6%	2059.4%
nondet.very-deep.32	3395.7%	1263.5%	4839.6%	1787.8%	—	17798.6%	100.0%	204.8%	532.2%	2021.6%	2021.6%	1064.3%	1066.2%	109.0%	18403.7%	18405.6%